It is not unique, because, for any arbitrary constant C, we get $\frac{d}{dx}[F(x)+C]=\frac{d}{dx}[F(x)]=f(x)$. That is, if $F(x)$ is an anti-derivative of $f(x)$, then the function $F(x)+C$ is also an anti-derivative of the same function $f(x)$. Note that all anti-derivatives of $f(x)$ differ by a constant only. The anti-derivative of $f(x)$ is usually called the **indefinite integral** of $f(x)$ with respect to x and is denoted by $\int f(x)dx$.

A well-known property of indefinite integral is its **linear property** :

$\int[\alpha f(x)+\beta g(x)]dx=\alpha\int f(x)dx+\beta\int g(x)dx$, where α and β are constants.

We list below some functions and their anti-derivatives (indefinite integrals):

Function $f(x)$	**Indefinite integral** $\int f(x)dx$
K, a constant	$Kx+C$
$(ax+b)^n$, where $a\neq 0$ and b are constants; and $n\neq -1$	$\frac{1}{a}\left[\frac{(ax+b)^{n+1}}{n+1}\right]+C$
$\frac{1}{ax+b}$, where $a\neq 0$ and b are constants	$\frac{1}{a}\log_e\lvert(ax+b)\rvert+C$
e^{ax}, where a is a non-zero constant	$\frac{e^{ax}}{a}+C$
$\sin(ax+b)$, where $a\neq 0$ and b are constants	$-\frac{\cos(ax+b)}{a}+C$
$\cos(ax+b)$, where $a\neq 0$ and b are constants	$\frac{\sin(ax+b)}{a}+C$
$\tan(ax+b)$, where $a\neq 0$ and b are constants	$\frac{1}{a}\log\lvert\sec(ax+b)\rvert+C$
$\cot(ax+b)$, where $a\neq 0$ and b are constants	$\frac{1}{a}\log\lvert\sin(ax+b)\rvert+C$
$\sec(ax+b)$, where $a\neq 0$ and b are constants	$\frac{1}{a}\log\lvert\sec(ax+b)+\tan(ax+b)\rvert+C$
$\operatorname{cosec}(ax+b)$, where $a\neq 0$ and b are constants	$-\frac{1}{a}\log\lvert\operatorname{cosec}(ax+b)-\cot(ax+b)\rvert+C$
$\frac{1}{a^2+x^2}$, where $a\neq 0$ is a constant	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$
$\frac{1}{a^2-x^2}$, where $a\neq 0$ is a constant	$\frac{1}{2a}\log_e\left\lvert\frac{a+x}{a-x}\right\rvert+C$
$\frac{1}{x^2-a^2}$, where $a\neq 0$ is a constant	$\frac{1}{2a}\log_e\left\lvert\frac{x-a}{x+a}\right\rvert+C$

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